

1. Match the layers—Link, Network, and Transport—with the guarantees that each layer could provide to higher layers.

Guarantee Layer	Layer
Best effort delivery	Network
Reliable delivery	Transport
In-order delivery	Transport
Byte-stream abstraction	Transport
Point-to-point link abstraction	Data link

2. Suppose that two network endpoints have a round-trip time of 100 milliseconds, and that the sender transmits five packets every round trip. What will be the sender's transmission rate for this round-trip time, assuming 1500-byte packets? Give your answer in bytes per second.

$$5 \text{ packets} \times 1500 \text{ bytes} = 7500 \text{ bytes per packet per } 100 \text{ ms}$$

$$1 \text{ ms} = 0.001 \text{ s}$$

$$100 \text{ ms} = 100 \times 0.001 \text{ s} = 0.1 \text{ s}$$

$$\text{Trate} = \frac{7500 \text{ bytes}}{0.1 \text{ bytes/s}} = 75000 \text{ bytes/s}$$

So the transmission rate is 75000 bytes per second

The subnet of Fig. 1-12(b) was designed to withstand a nuclear war. How many bombs would it take to partition the nodes into two disconnected sets? Assume that any bomb wipes out a node and all of the links connected to it.

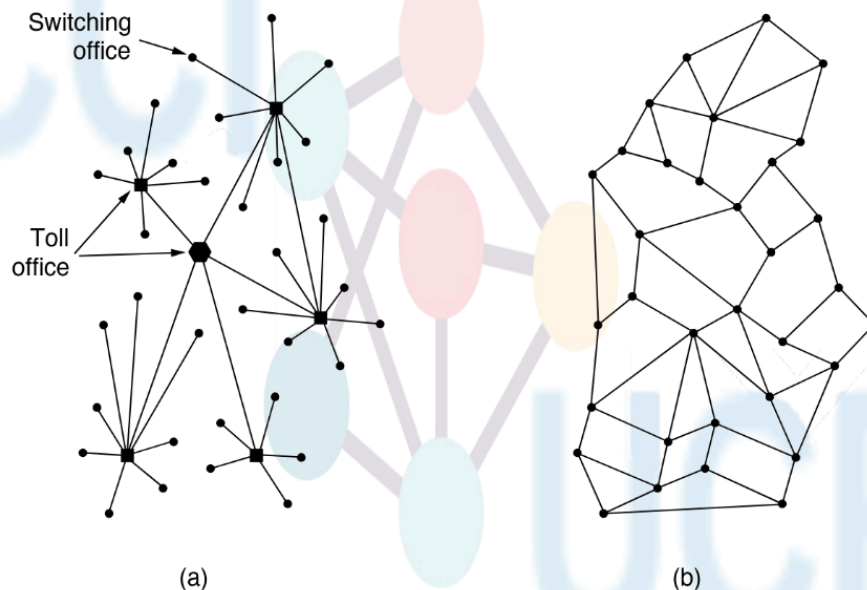
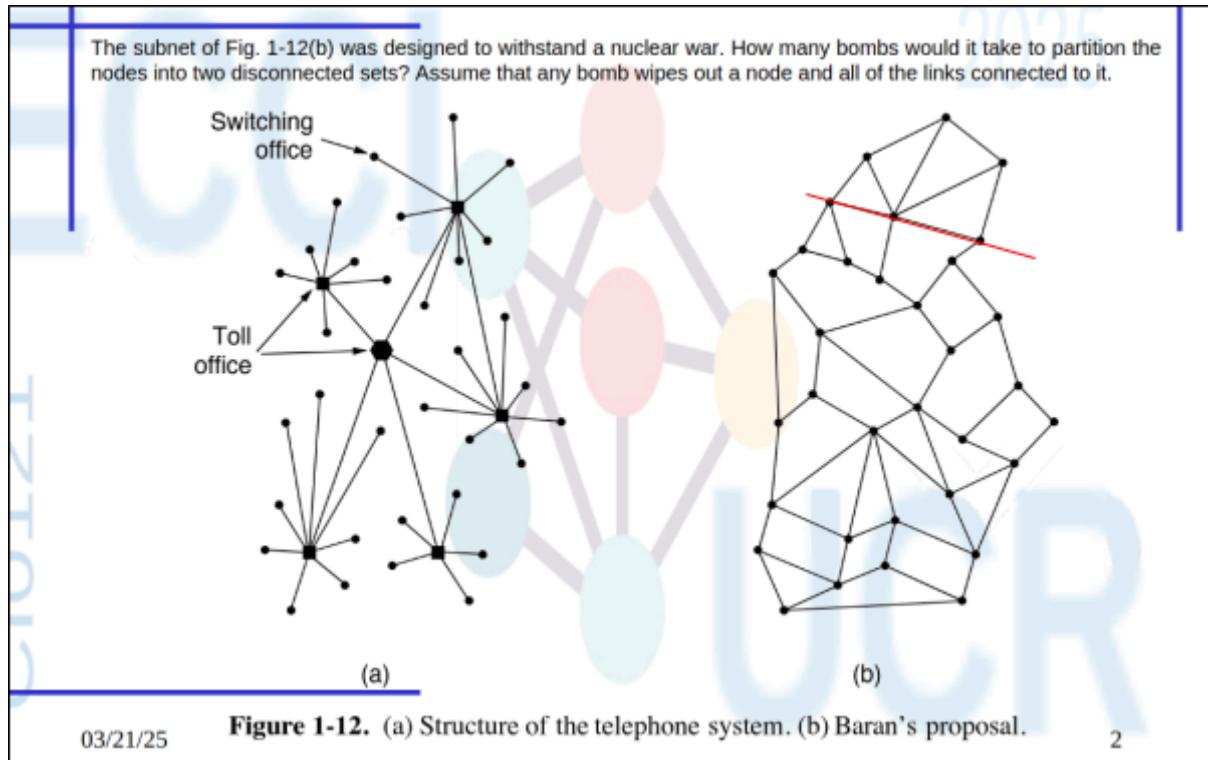


Figure 1-12. (a) Structure of the telephone system. (b) Baran's proposal.

Note que hay algunos nodos con solamente 3 conexiones, si se explotan los tres nodos de arriba se podrían separar en dos conjuntos, sin que puedan tener comunicación los nodos, así la cantidad de bombas a explotar serían 3 para que quedarán en dos conjuntos separados, adjunto una imagen con un ejemplo de ello.



La línea roja da un ejemplo de los nodos que se necesitan explotar con bombas para separar esa red.